

VesperGrid

Critical Infrastructure Operational Twin on AMD MI300X

A source-linked decision-support model for industrial safety incidents.

Multimodal evidence in. Candidate plans, with audit trail, out.

Track 3 · Qwen Challenge · HF Special Prize · Build-in-Public

The 18-minute problem

A port operator gets four signals at once:

- one **drone keyframe**
- one **gate-camera clip**
- one **wind-sensor strip**
- one **operator radio note**

They contradict each other. There is no single source of truth.
Decision pressure is rising. A wrong route can cost lives.

What VesperGrid does

“Turn **fragmented multimodal evidence** into a **source-linked operational twin**, with uncertainty named, not buried.”

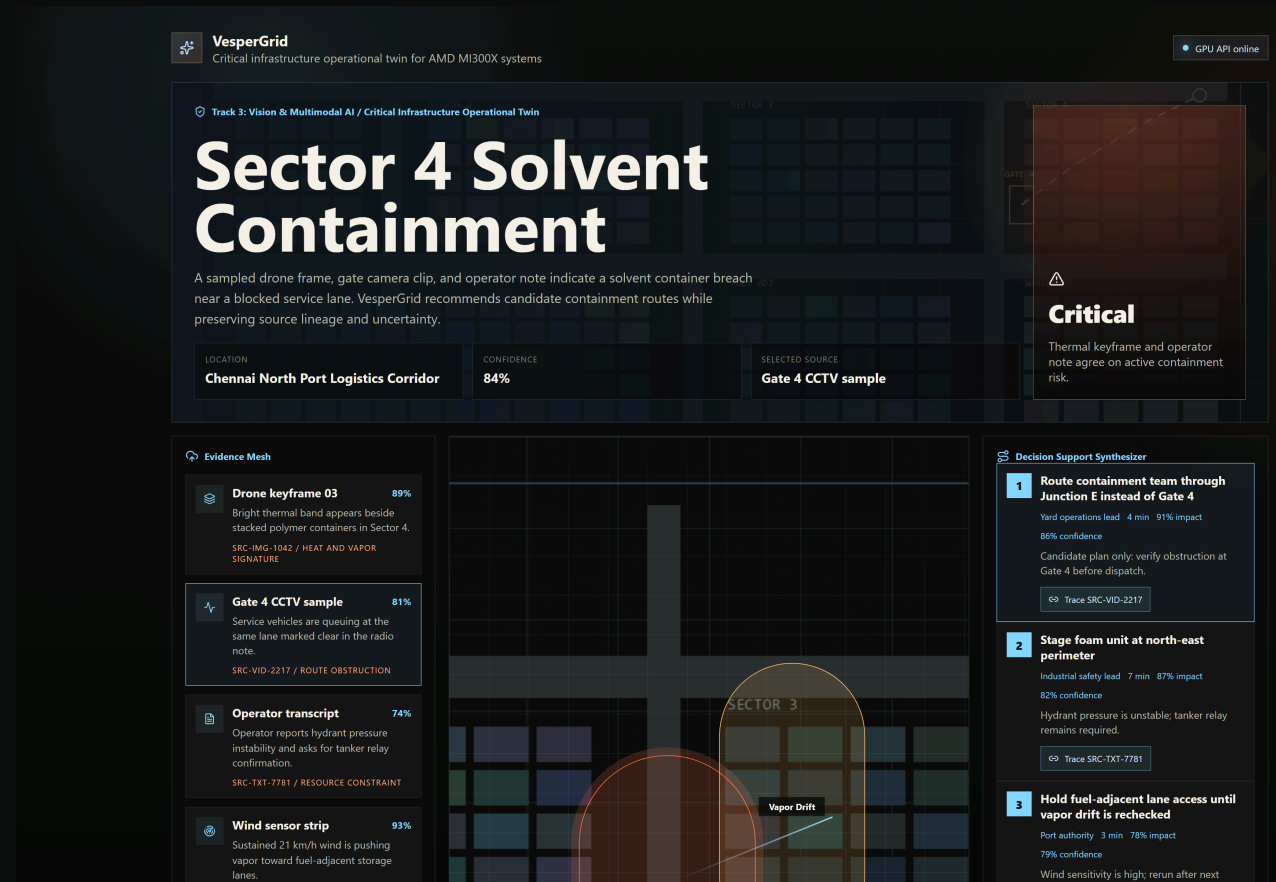
```
evidence pack → async ingest → Qwen-VL on MI300X  
    → normalized graph → candidate plan + uncertainty ledger  
    → every claim links back to a SRC-* UUID
```

The console is the operator's read-only view of the evidence-to-decision chain.

Live demo — operations console

Hero band: incident,
severity, location,
confidence

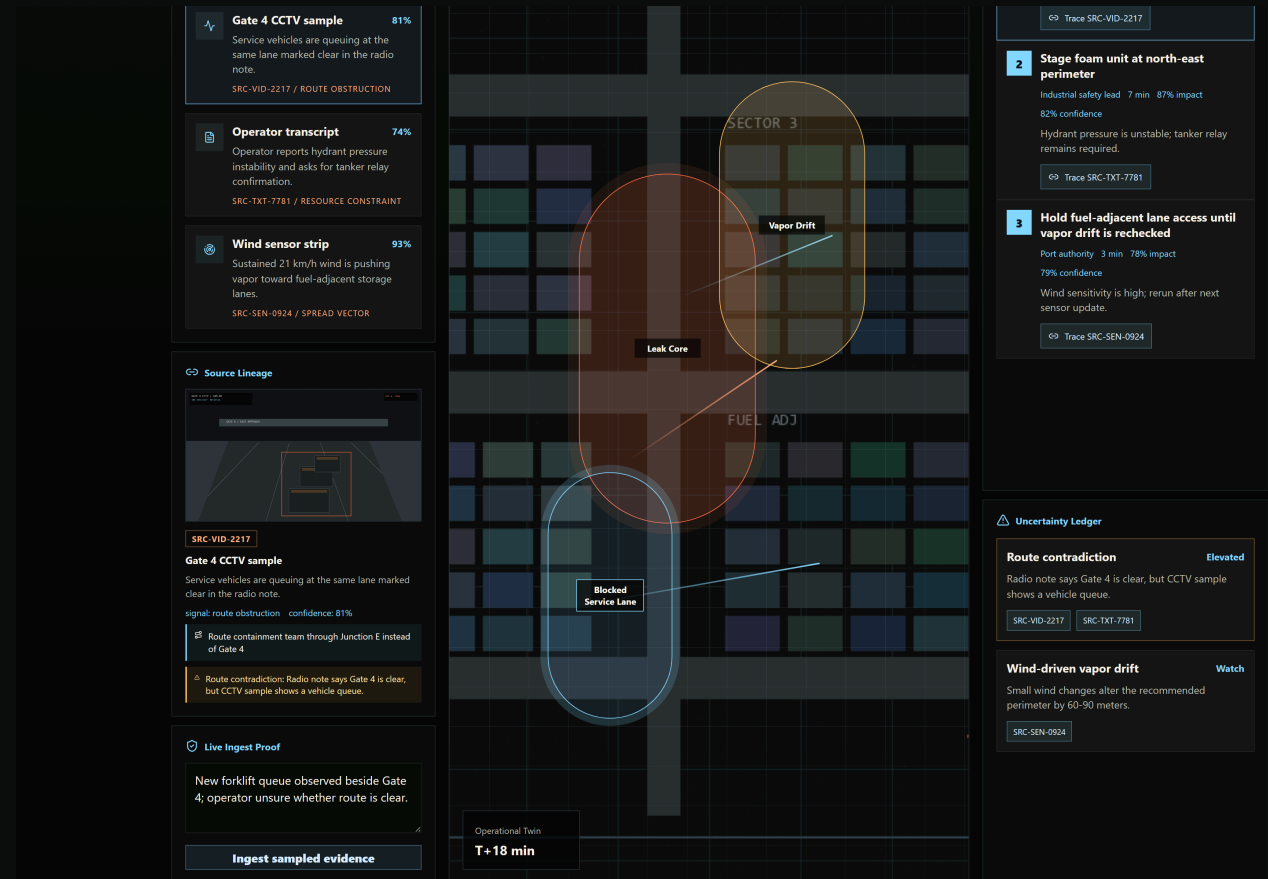
Evidence Mesh (left):
four heterogeneous
sources, each with a



Source lineage — the wow moment

Click `SRC-VID-2217`.

The same UUID surfaces:



Async pipeline — no synchronous trap

```
POST /api/ingest          → { job_id }  
GET  /api/ingest/{id}/events (SSE)
```

```
queued → sampling → parsing → normalizing → synthesizing → complete
```

Per-stage SSE events drive a real progress bar in the console.

If the GPU backend fails mid-pipeline, the API soft-degrades to a deterministic

synthesizer. The demo never breaks. Schemas are Pydantic-

Why AMD MI300X

Resource	Value
GPU	1 × AMD Instinct MI300X
GPU memory	192 GB VRAM
Model	Qwen2.5-VL-7B-Instruct via vLLM ROCm
Tensor parallel	TP=1 (single GPU, zero NCCL/RCCL surface)
Multi-image budget	5 keyframes per prompt, resident

Honesty

What is **GPU-resident** on the MI300X:

→ Qwen-VL multimodal inference (paged attention, KV cache, multi-image decode)

What is intentionally **CPU-vectorized** for the MVP:

→ bounded route/hazard scoring

→ in-memory entity graph

The Uncertainty Ledger is **first-class**, not an afterthought. Every

Business value

Buyers feel the pain in cost-per-minute of downtime:

- **Ports & logistics yards** — vessel queuing, gate dwell, dangerous-goods routing
- **Energy & petrochemical** — perimeter incidents, vapor drift, evacuation paths
- **Heavy industry** — forklift collisions, contained-spill protocols

Roadmap (post-hackathon)

Submission map

Asset	Location
Code repository	<code><github-url></code>
Live app (HF Space)	<code><huggingface.co/spaces/...></code>
AMD Cloud deploy	<code>scripts/bootstrap_amd_cloud.sh</code> (idempotent, single MI300X)
Demo evidence pack	<code>demo/sector4/</code> (synthetic, license-clean)